Haya Mahmoud

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Professional Experience

Full Stack Software Developer Co-op *Toronto, ON / Remote*
FreshBooks *Sept 2026 – Present*

- Ship full stack features across multiple FreshBooks' financial platform, contributing to weekly production releases supporting **30M+** users across North America and Europe.
- Develop and maintain frontend and backend features across **4+** product surfaces, integrating third-party APIs and implementing automated tests to improve code quality and release reliability.
- Refactor **3** backend modules and **6+** integration components toward a **hexagonal architecture**, separating domain logic from external dependencies to improve maintainability, testability, and extensibility for future integrations.
- Optimize backend integrations by upgrading **14+** API endpoints across **4+** payroll workflows, reducing average response latency by **8%**.

Web Applications Developer Co-op *Mississauga, ON / Hybrid*
Kenna *May 2026 – Aug 2026*

- Developed and maintained full stack web applications for enterprise clients, including campaign sites, internal tools, and client-facing workflows supporting **20,000+** active users across the U.S. and Canada.
- Optimized features using modern technologies, improving performance by **30%** and strengthening security.
- Improved deployment reliability through code reviews and testing, reducing rollbacks by **15%**.

Projects

PosturePal *May 2026*
React, Vite, Tailwind CSS, MediaPipe Pose, Firebase Auth, Firestore github.com/hayamahmoudd/PosturePal

- Developed a real-time posture coaching application using MediaPipe Pose to track **33** body landmarks and calculate posture scores from webcam input.
- Implemented personalized calibration, configurable thresholds, and real-time alerts to detect and flag poor posture
- Integrated **Firebase Authentication** and **Firestore** to support **4+** features, including session history, posture analytics, leaderboards, and social interactions.

Pray360 *Jan 2026*
TypeScript, React, MapTiler, Fastify, PostgreSQL, Docker github.com/hayamahmoudd/Pray360

- Built a platform for discovering and reviewing prayer spaces using **React, Fastify, and PostgreSQL.**
- Engineered RESTful APIs with input validation and structured error handling, containerizing backend services with **Docker** for consistent deployment.
- Secured **3rd place** in the Open Innovation category at a competition with **300+ participants**.

ShelterMap Toronto *Nov 2025*
Python, Next.js, React-Leaflet, Google Cloud, BigQuery, Docker github.com/hayamahmoudd/ShelterMap

- Engineered a Python data pipeline to process **500,000+** records, identifying high-priority shelter locations across the Greater Toronto Area through an interactive **Next.js/React** map.
- Leveraged **BigQuery** for large-scale data analysis and deployed containerized services using **Docker** and **GCP**.
- Earned a **Best Use of Google Cloud** nomination.

Education

Bachelor of Computing, Software Engineering (Co-op) *Sept 2024 – Expected Apr 2029*
University of Guelph

- Area of Emphasis: Artificial Intelligence

Technical Skills

Languages: Java, Python, C, C#, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks: React, Next.js, Node.js, Fastify, Vite, Tailwind CSS, Firebase

Cloud and Tools: Google Cloud Platform, BigQuery, Docker, AWS, Git, Linux/Unix, Bash, Jenkins, Jira, VS Code, Valgrind

Concepts: Data Structures and Algorithms, OOP, REST APIs, Relational Databases, CI/CD, Unit Testing, Agile/Scrum

Awards & Achievements

University of Guelph's CEPS Dean's Honours List (4x) *Sep 2024 – May 2026*

- Recognized for achieving an average of **80%+** per semester.